

CKS121/CSE 242

Software Requirements

Chapter 1: Introduction &
Fundamental of Requirements
Engineering (RE)

Learning outcomes

The students

- understand the importance of requirements engineering for the project success.
- learn the definitions of requirements engineering and related terms.
- understand the three dimensions of requirements engineering and the requirements engineering process within these dimensions.
- learn the goals of requirements engineering.
- learn different types of requirements.

Structure of this Lecture

- Importance of Requirements Engineering
- Three Dimensions of Requirements Engineering
- Three Types of Requirements



How would you describe the term “requirements” to your grandfather?

Share your answer in the chat box.



Requirement Engineering : What is it ?

Definition :

1. A **condition** or **capability** needed by the user to **solve a problem or achieve an objective**
2. A condition or capability that must be met or possessed by a system or system component to **satisfy a contract, standard, specification, or other formally imposed documents.**
3. A documented representation of a condition or capability as in (1) or (2)

No Silver Bullet

“The hardest single part of building a software system is deciding precisely what to build.

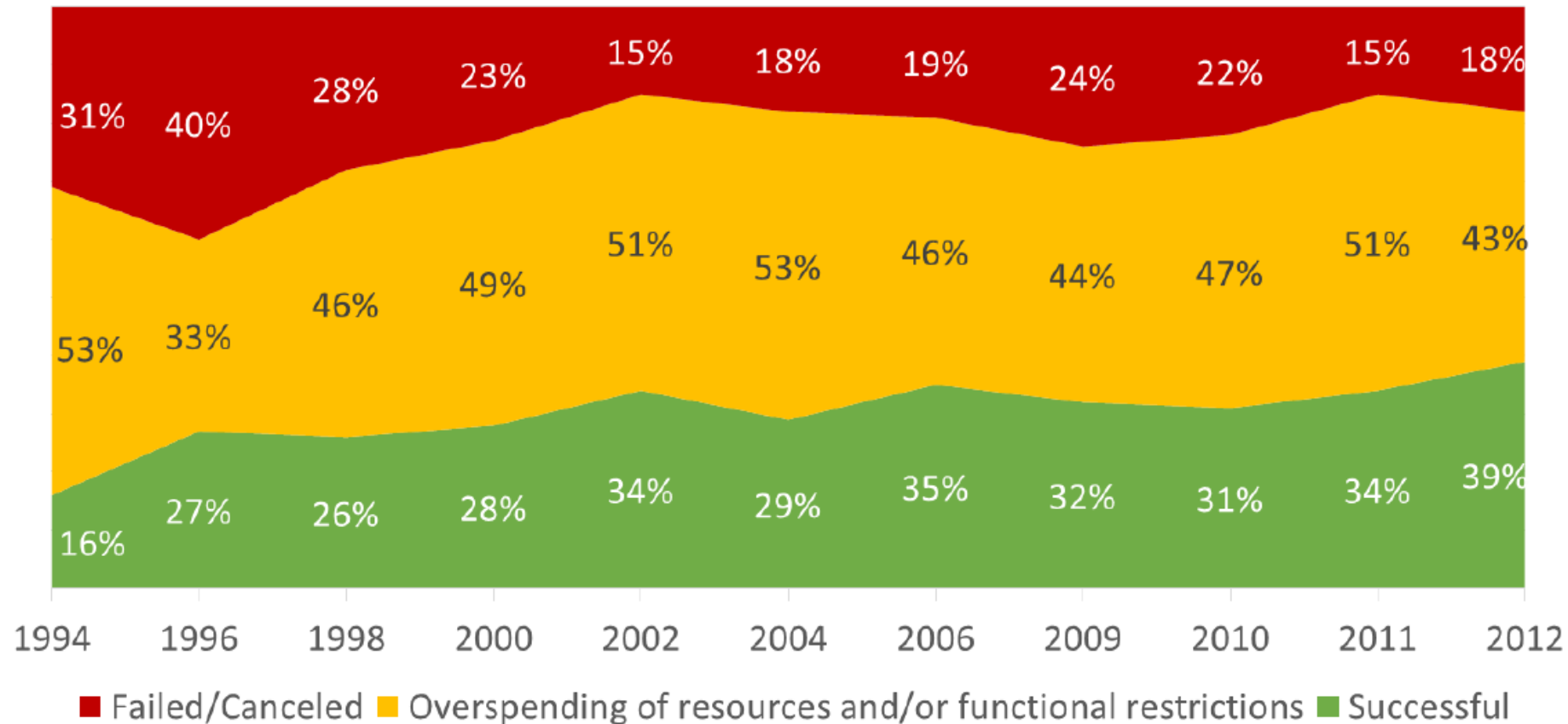
No other part of the conceptual work is as difficult as establishing the detailed technical requirements, including all the interfaces to people, to machines, and to other software systems.

No other part of the work so cripples the resulting system if done wrong. No other part is more difficult to rectify later.”

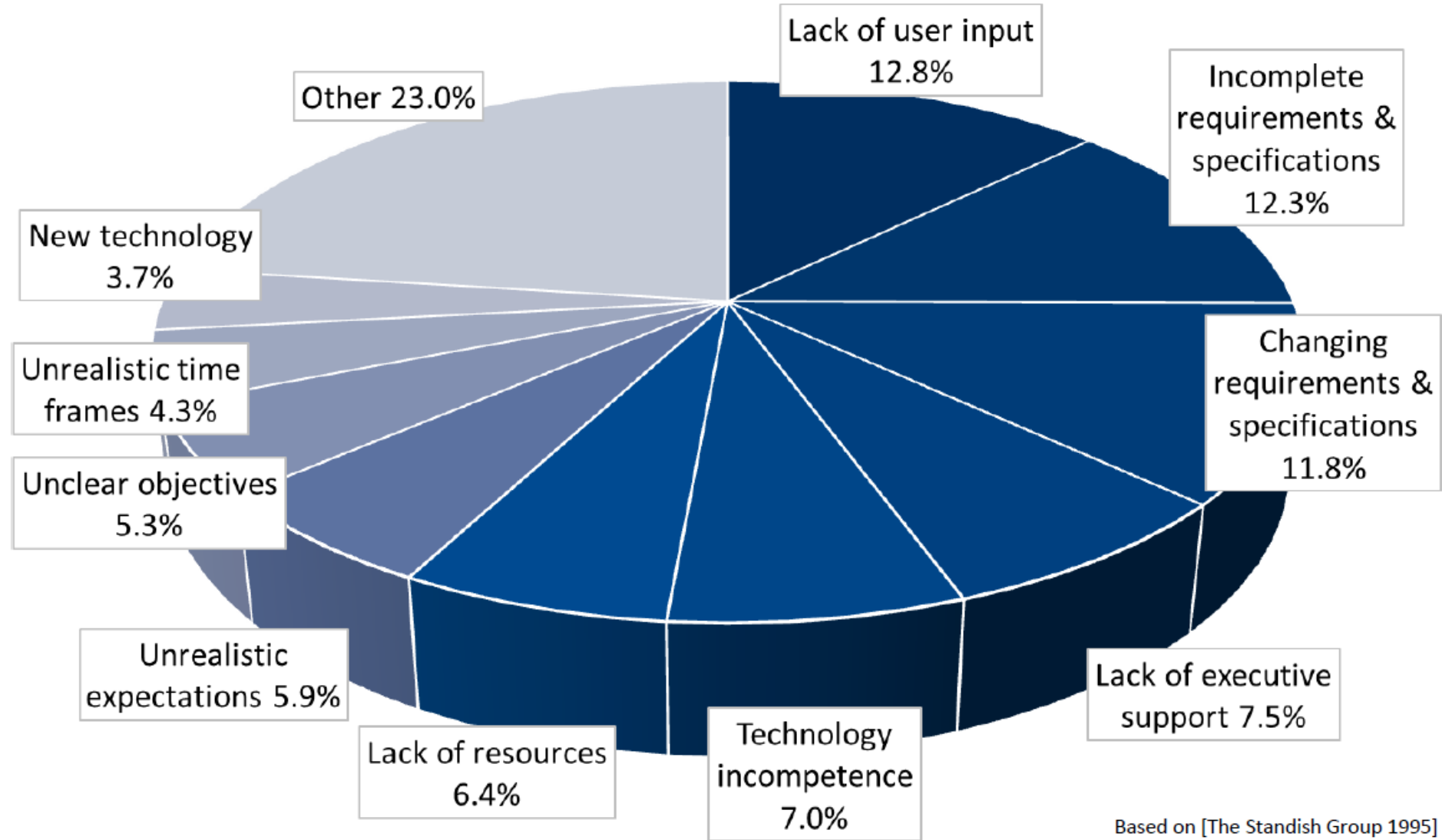
[Brooks 1987]

CHAOS Report – Project Success and Failure

In average only **30%** of the projects were successful, **23%** were cancelled, and **47%** were completed with resource overspend and functional restrictions.

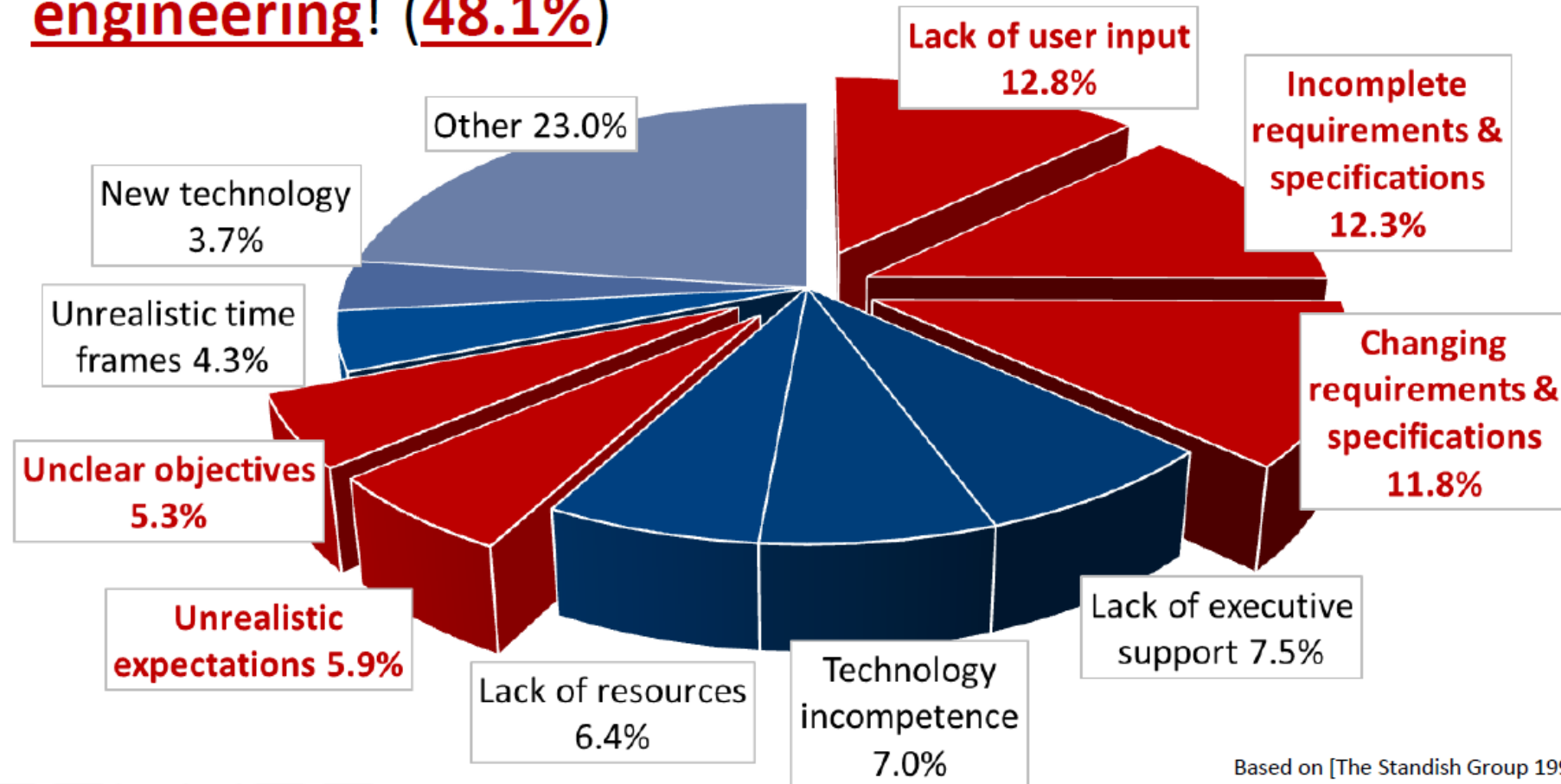


CHAOS Report – Reasons for Resource Overspend and Functional Restrictions

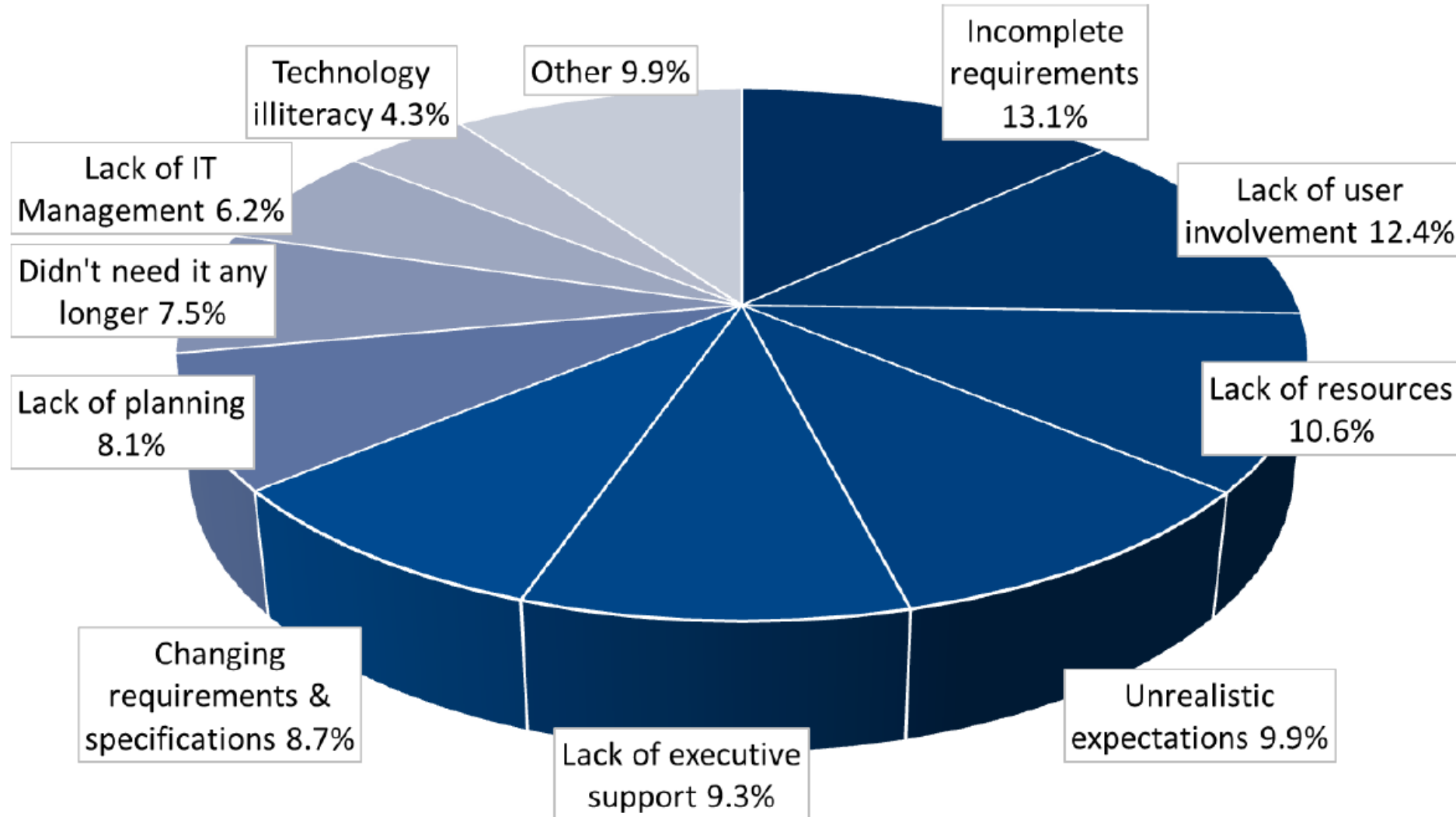


CHAOS Report – Reasons for Resource Overspend and Functional Restrictions

Reasons related to insufficient and poor requirements engineering! (**48.1%**)

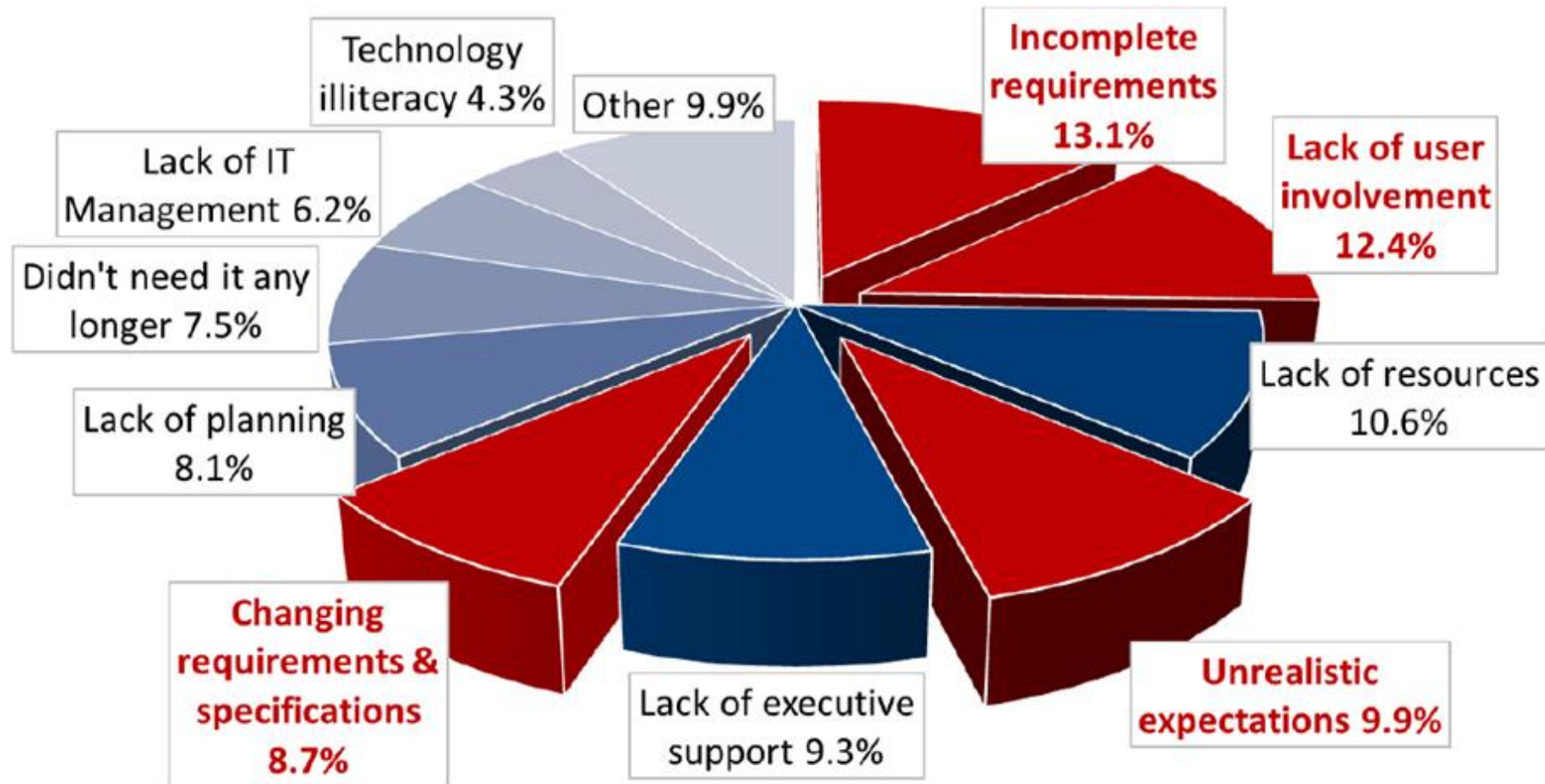


CHAOS Report – Reasons for Projects Cancelled



CHAOS Report – Reasons for Projects Cancelled

Reasons related to insufficient and poor requirements engineering! (44.1%)



Requirements Defects

Effort required for correcting requirements defects:

- Defect found after delivery (small & non-critical projects)
→ increase of effort required for correction by factor 5!
- Defect found after delivery (large & critical projects)
→ increase of effort required for correction by factor 100!



So, what is your definition of a successful project?

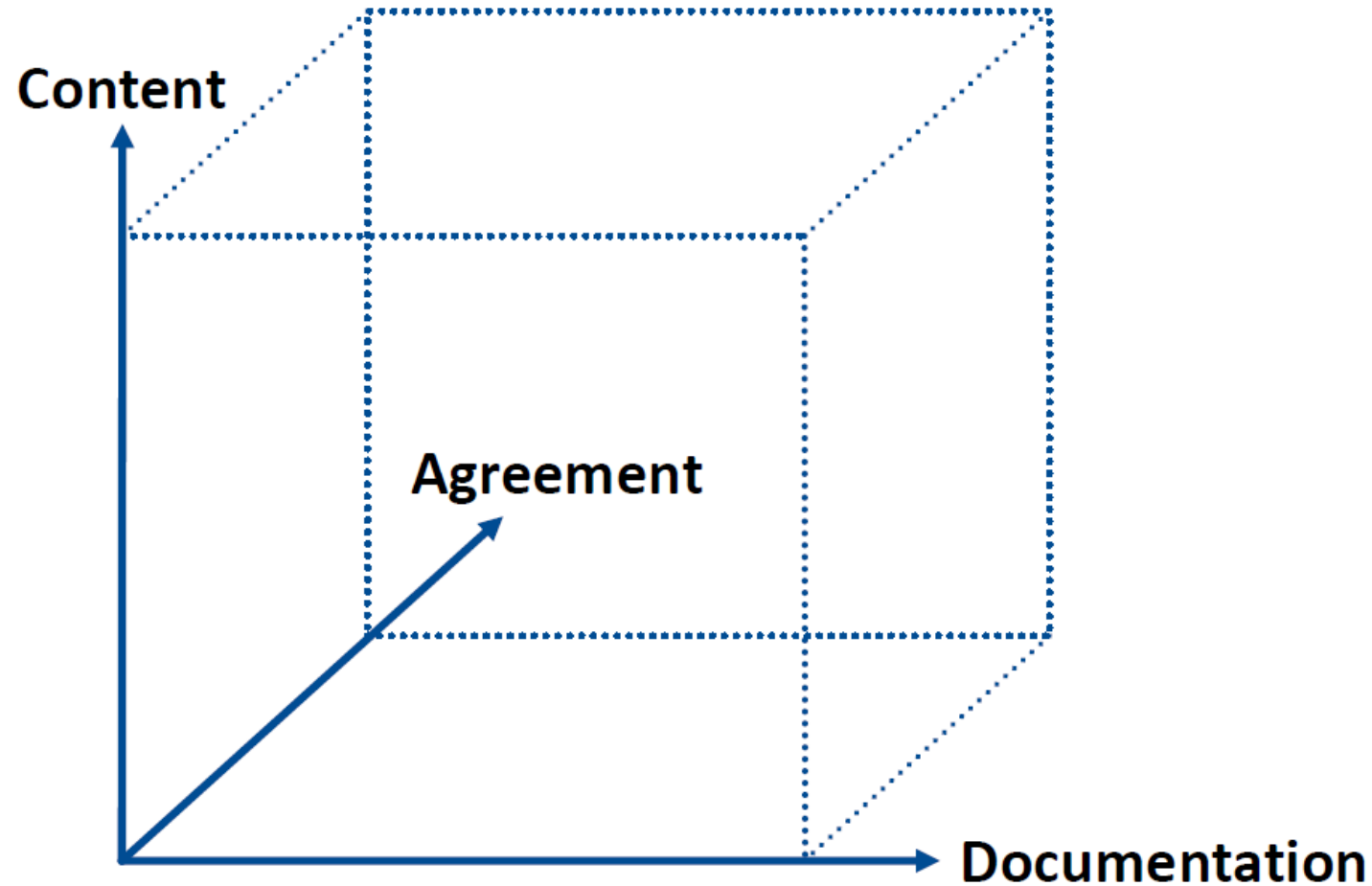
Share your answer in the chat box.

Structure of this Lecture

- Importance of Requirements Engineering
- **Three Dimensions of Requirements Engineering**
- Three Types of Requirements



Three Dimensions of Requirements Engineering



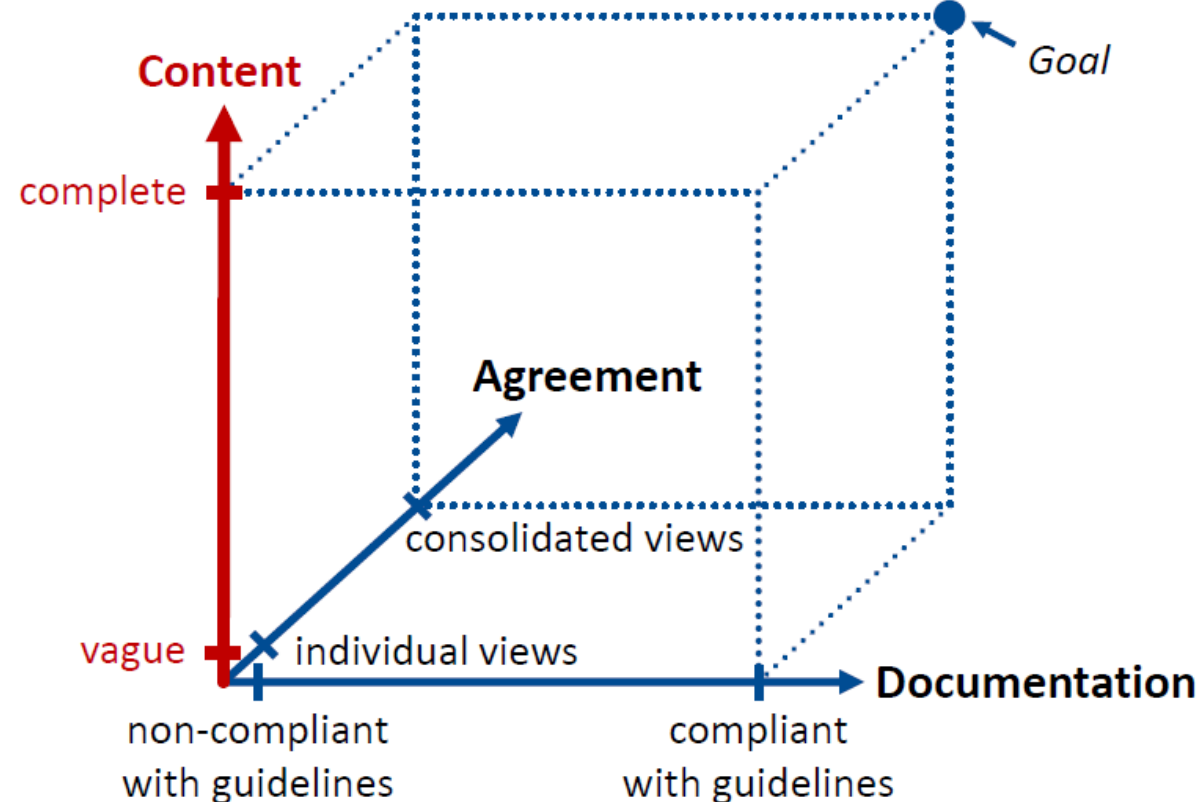
Content Dimension

- The content dimension deals with the understanding of the system requirements.
- At the beginning of the requirements engineering process only a few requirements are known and their understanding is typically vague.



At the end:

All requirements shall be explicitly known and each requirement shall be understood at the required level of detail.

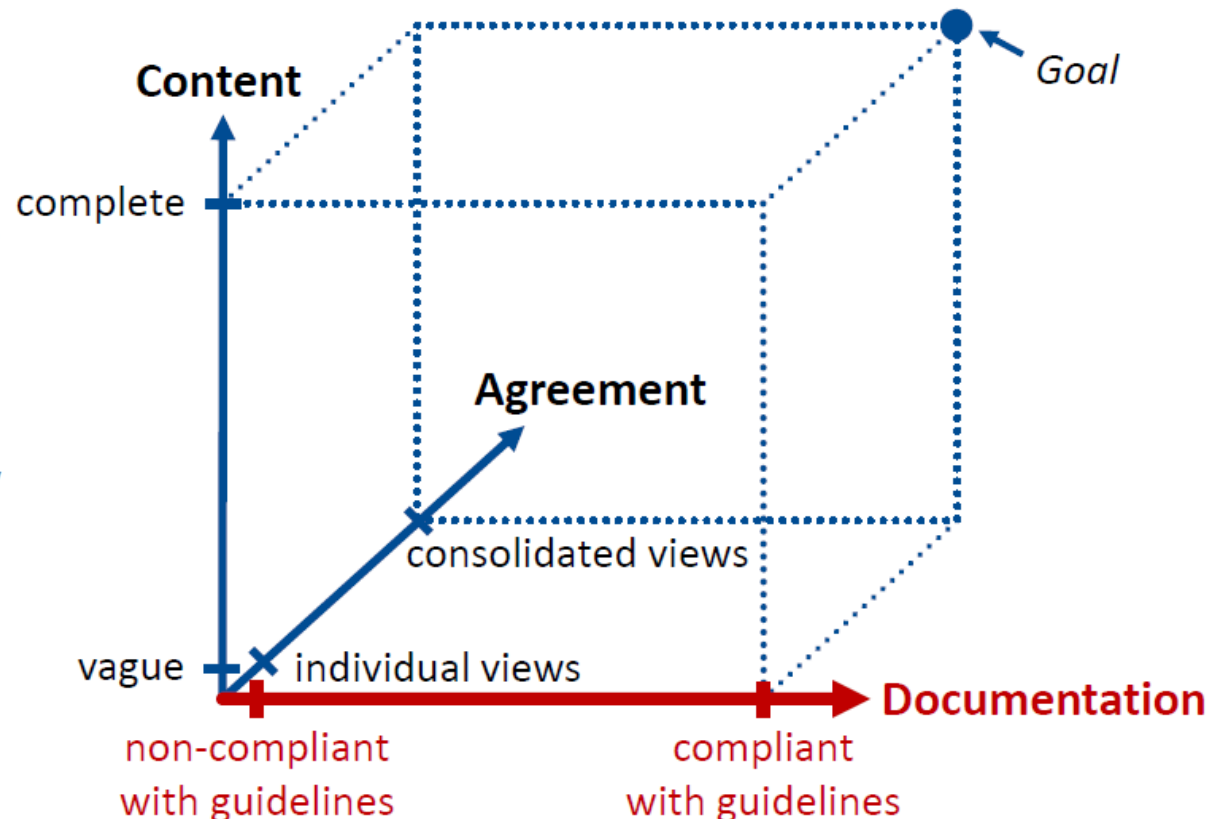


Documentation Dimension

- The documentation dimension deals with documenting and specifying the system requirements using different and appropriate documentation and specification formats.
- At the beginning information is typically documented non-compliant with documentation / specification guidelines, e.g., either as a note, sketch, statement in a minute, hand drawing, ...



At the end:
All requirements should be documented / specified in compliance with the relevant documentation / specification guidelines and formats.

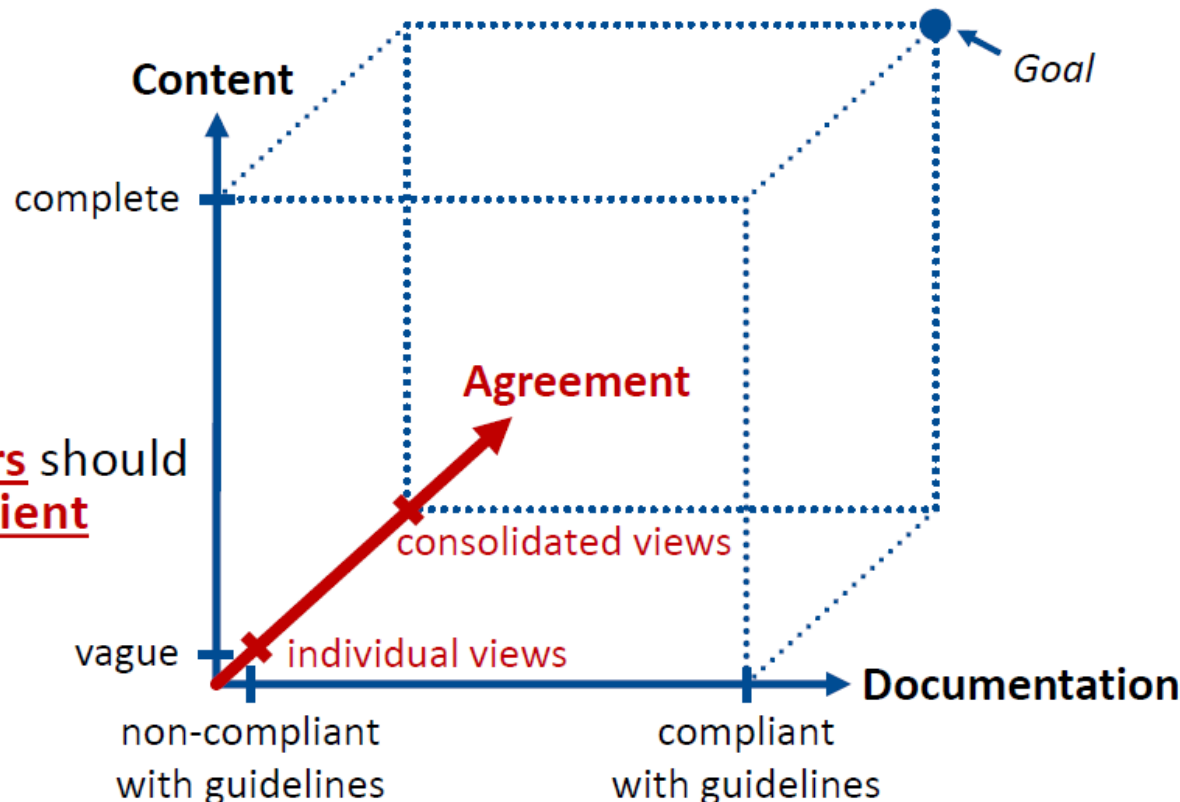


Agreement Dimension

- The agreement dimension deals with the level of agreement achieved between the relevant stakeholders about the known requirements.
- At the beginning stakeholders typically have different, conflicting views about the goals and requirements of the system.
- Conflicts between stakeholders about requirements have to be detected and resolved as early as possible.

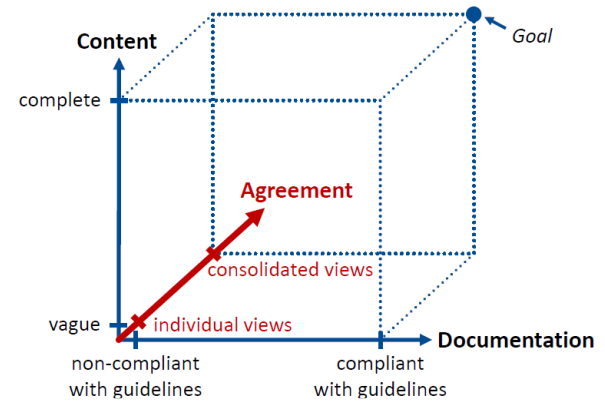


At the end:
The relevant stakeholders should have established a sufficient agreement about the system requirements.



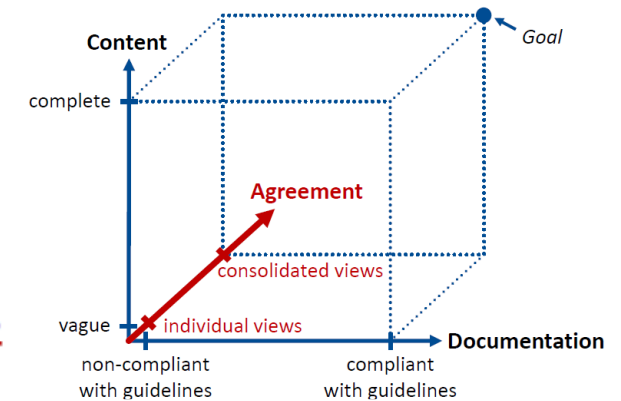
Orthogonality of Content and Agreement

- A complete understanding of a requirement does not imply a sufficient agreement between the stakeholders about this requirement!
- An agreement between stakeholders about a requirement does not imply a complete understanding of this requirement!
- Progress
 - in the agreement dimension (e.g., a creative solution for a conflict on requirements) can lead to new requirements which are not well understood (content dimension).
 - in the content dimension (e.g., by eliciting new requirements) may lead to new conflicts between stakeholders, or may uncover existing conflicts (agreement dimension).



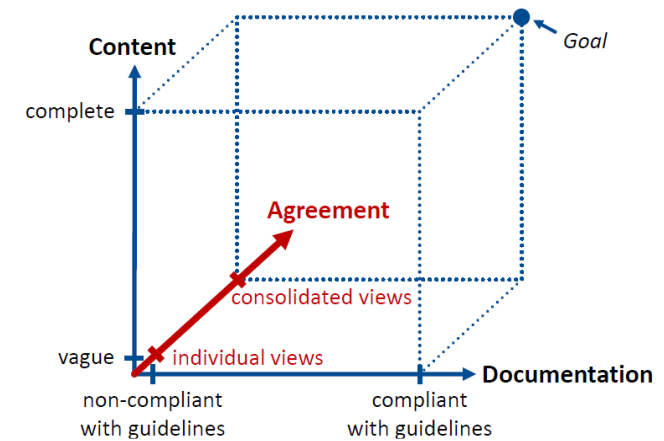
Orthogonality of Documentation and Content

- Compliance of a requirement to the documentation guidelines does not imply a complete understanding of the requirement!
- A complete understanding of a requirement does not imply the compliance of the documentation of the requirement with the documentation guidelines!
- Progress
 - in the documentation dimension (e.g., the formalization of a requirement according to the guidelines) can reveal some gaps within the content dimension.
 - in the content dimension (e.g., a new elicited requirement) can lead to a drawback in the documentation dimension since, most likely, the new requirement is not directly documented according to the documentation guidelines.



Orthogonality of Agreement and Documentation

- An agreement between stakeholders about a requirement does not imply compliance of the documentation of the requirement to the documentation guidelines!
- Compliance of a requirement with the documentation guidelines does not imply a sufficient agreement of the stakeholders about this requirement!
- Progress
 - in the documentation dimension (e.g., the documentation of a requirement according to the guidelines) can surface a conflict about the requirement (e.g., due to better comprehension of the requirement by a stakeholder).
 - in the agreement dimension (e.g., the stakeholders solve a conflict by agreeing on a new alternative requirement) can lead to a drawback in the documentation dimension, if the new documented requirement is initially non-compliant with documentation guidelines.





How would you describe the term “stakeholder” to your sister/brother?

Share your answer in the chat box.

Definition: Stakeholder

The stakeholder of a system is :

- A person, group of people or an organisation
- Who has **directly or indirectly**
- **Influence** on the requirements of the regarded system

A stakeholder can influence not only the result, but also the project life cycle.



So, what is the goal of
Requirement
Engineering (RE)?

(Goal of) Requirements Engineering

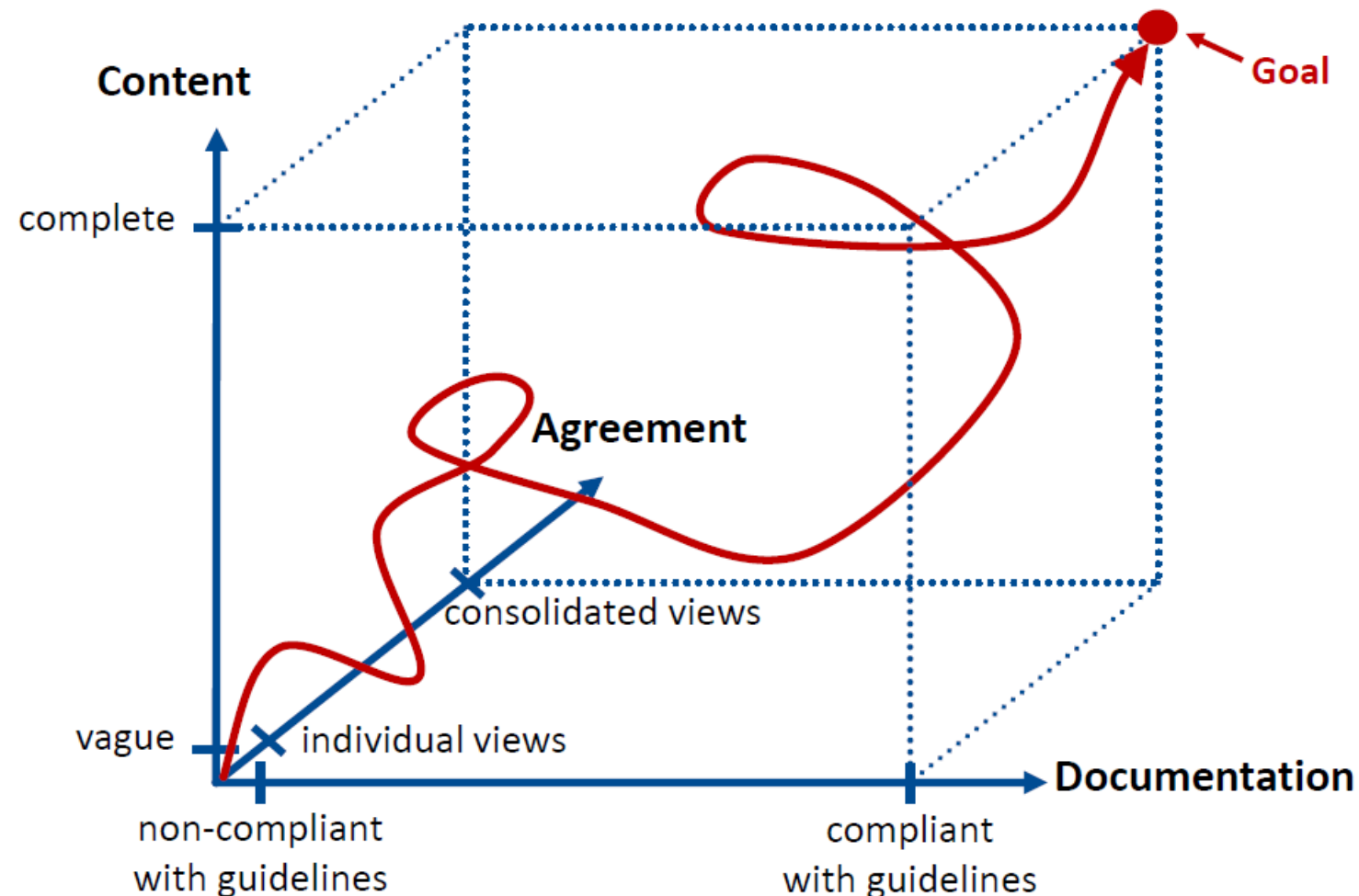
- D** Requirements engineering (RE) is a cooperative, iterative and incremental process which aims at ensuring that:
- (1) All relevant requirements are explicitly known and understood at the required level of detail.
 - (2) A sufficient agreement about the system requirements is achieved between the stakeholders involved.
 - (3) All requirements are documented and specified in compliance with the relevant documentation/specification guidelines.

content

agreement

document
ation

Requirements Engineering Process within the Three Dimensions



Structure of this Lecture

- Importance of Requirements Engineering
- Three Dimensions of Requirements Engineering
- **Three Types of Requirements**



The background of the slide features a light blue wall with several white ladders leaning against it. One ladder is positioned centrally and is taller than the others, while several shorter ladders are placed around it, creating a sense of depth and repetition.

Recall of requirement
definition...

Requirement

D

A requirement is

- (1) a condition or capability needed by a user to solve a problem or achieve an objective.
- (2) a condition or capability that shall be met or possessed by a system, system component, product or service to satisfy a contract, an agreement, standard, specification or other formally imposed documents.
- (3) a documented representation of a condition or capability as in (1) or (2).

Requirements include the quantified and documented needs, wants and expectations of stakeholders.



What are the 3 types of requirements?

Three Types of Requirements

Functional Requirements (1)



Functional
Requirements



Quality
Requirements



Constraints

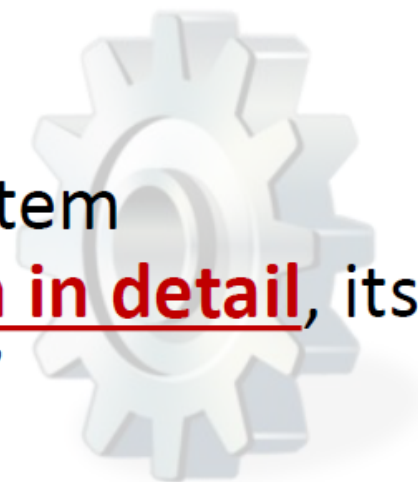
Functional Requirements (2)



“These [functional requirements] are statements of services the system should provide, how the system should react to particular inputs and how the system should behave in particular situations.

In some cases, the functional requirements may also state what the system should not do. [...]

When expressed as user requirements, the requirements are usually described in a fairly abstract way. However, functional system requirements describe the system function in detail, its inputs and outputs, exceptions and so on.”



Functional Requirements (3)



-  R-4: The house information system shall generate monthly statements of allowed and denied accesses.
-  R-6: If a sensor detects a damage of the window, the system shall inform the security company.
-  R-9: If a correct PIN is entered at the keyboard of the access system, the system shall remove the door lock and shall record access date, time and the PIN entered.

Quality Requirements (1)



Functional
Requirements



Quality
Requirements



Constraints

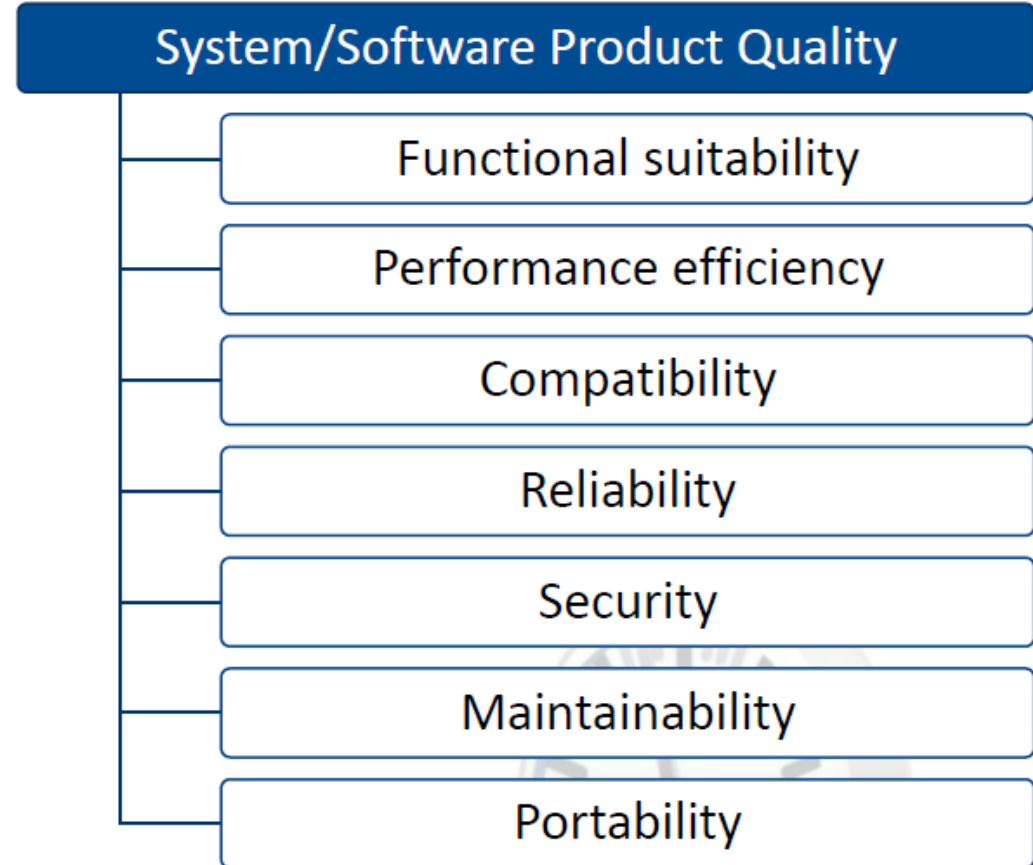
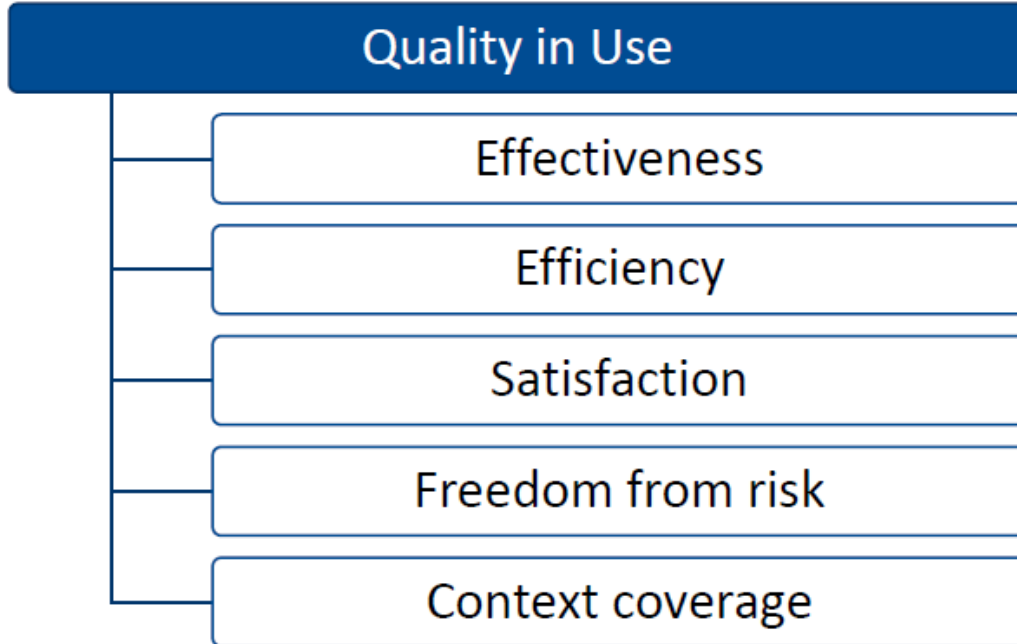
Quality Requirements (2)



A quality requirement defines a quality property for the entire system, for a system component, for a service or for a function.



Quality Requirements (3)



Quality in Use

Quality Requirement	Description
	Accuracy and completeness with which users achieve specified goals.
	Resources expended in relation to the accuracy and completeness with which users achieve goals.
	Degree to which user needs are satisfied when a product or system is used in a specified context of use.
	Degree to which a product or system mitigates the potential risk to economic status, human life, health or the environment.
	Degree to which a product or system can be used with effectiveness, efficiency, freedom from risk and satisfaction in both contexts of use and in contexts beyond those initially explicitly identified.

Quality in Use

Effectiveness

Efficiency

Satisfaction

Freedom from risk

Context coverage

Three Types of Requirements

System/Software Product Quality (1)

Quality Requirement	Description
	Degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.
	Performance relative to the amount of resources (e.g., other software products, hardware configuration, etc.) used under stated conditions.
	Degree to which a product, system or component can exchange information with other products, systems or components and/or perform its required functions, while sharing the same hardware or software environment.

System/Software Product Quality

Functional suitability

Performance efficiency

Compatibility

Reliability

Security

Maintainability

Portability

System/Software Product Quality (2)

Quality Requirement	Description
	Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.
	Degree to which a product or system protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorisation.
	Degree of effectiveness and efficiency with which a product or system can be modified by the intended maintainers.
	Degree of effectiveness and efficiency with which a system, product or component can be transferred from one hardware, software or other operation or usage environment to another.

System/Software Product Quality

Functional suitability

Performance efficiency

Compatibility

Reliability

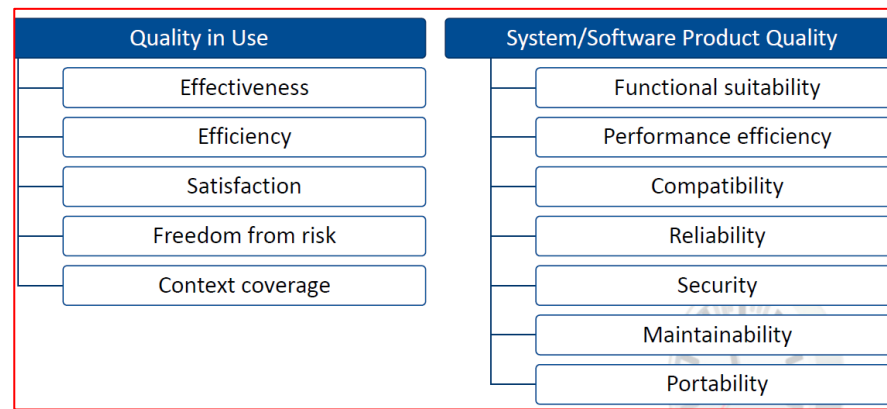
Security

Maintainability

Portability

Three Types of Requirements

Quality Requirements



Quality Requirement	Type of Quality
 R-12: The user interface shall be easy to use for the house owner.	
 R-15: The release of the locking mechanism shall take 0.8 seconds at most.	
 R-17: The user password stored in the system shall be protected against unauthorized access.	

Constraints (1)



Functional
Requirements



Quality
Requirements



Constraints

Constraints (2)

D A constraint is an organizational or technological requirement which restricts the way the system shall be developed.

Based on [Robertson and Robertson 2006]

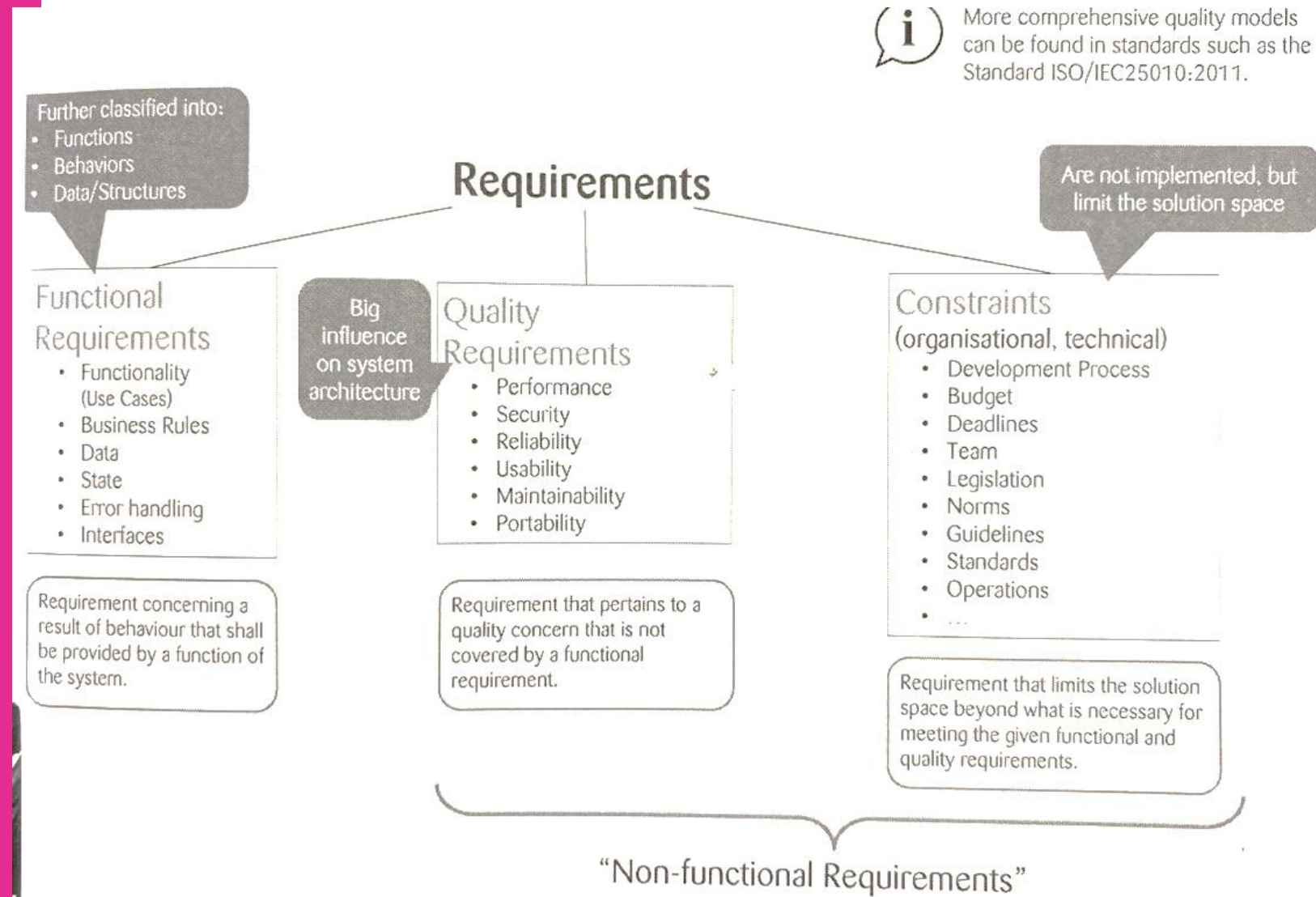


Constraints (3)



Constraint	Type of Constraint
 C-9: The user interface shall not contain symbols or graphics abusive for any culture.	culture
 C-16: The effort for system development shall not exceed 480 person months.	organisational/ project
 C-36: The electronic control unit in the vehicle interior shall work at temperatures from -10 to +50 degrees Celsius.	physical
 C-41: The system shall process personal data in compliance with the EU's Data Protection Directive 95/46/EC.	legal

Summary of Requirement Types



Summary

- Requirements engineering is a critical success factor for software-intensive systems.
- Definition of requirements engineering and requirement.
- Requirements engineering has three main goals characterised by the three dimensions of requirements engineering (content, documentation and agreement).
- Three types of requirements:
Functional requirements, quality requirements and constraints.

Questions for Self-Assessment

- How are the terms “requirements engineering” and “requirements” defined?
- Name the three goals of requirements engineering!
- How are the three dimensions interrelated?
- What is the goal of each dimension?
- Which types of requirements do you know?

Questions ???

This Lecture Material was produced by

Prof. Dr. Klaus Pohl

Dr. Thorsten Weyer

Torsten Bandyszak

Philipp Bohn

Jennifer Brings

Johanna Buchner

Marian Daun

Felix Föcker

Andrea Salmon

Philipp Schmidt

Bastian Tenbergen

**paluno – The Ruhr Institute for Software Technology
University of Duisburg-Essen
Germany**

UNIVERSITÄT
DUISBURG
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PALUNO
The Ruhr Institute for Software Technology
<https://www.paluno.uni-due.de>